SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I Sakthi Priya J 192411110certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAKTHI PRIYA J

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

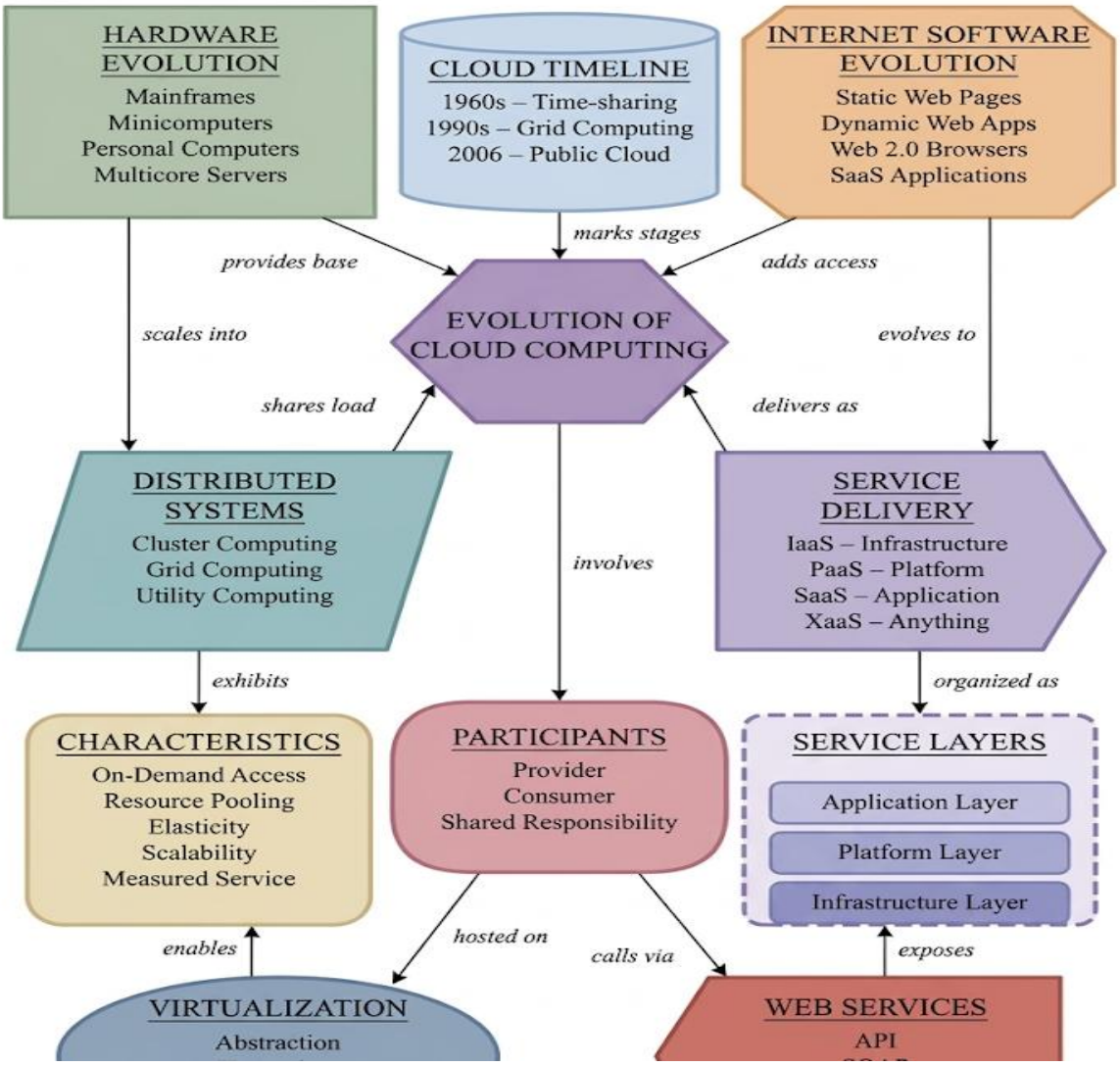
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

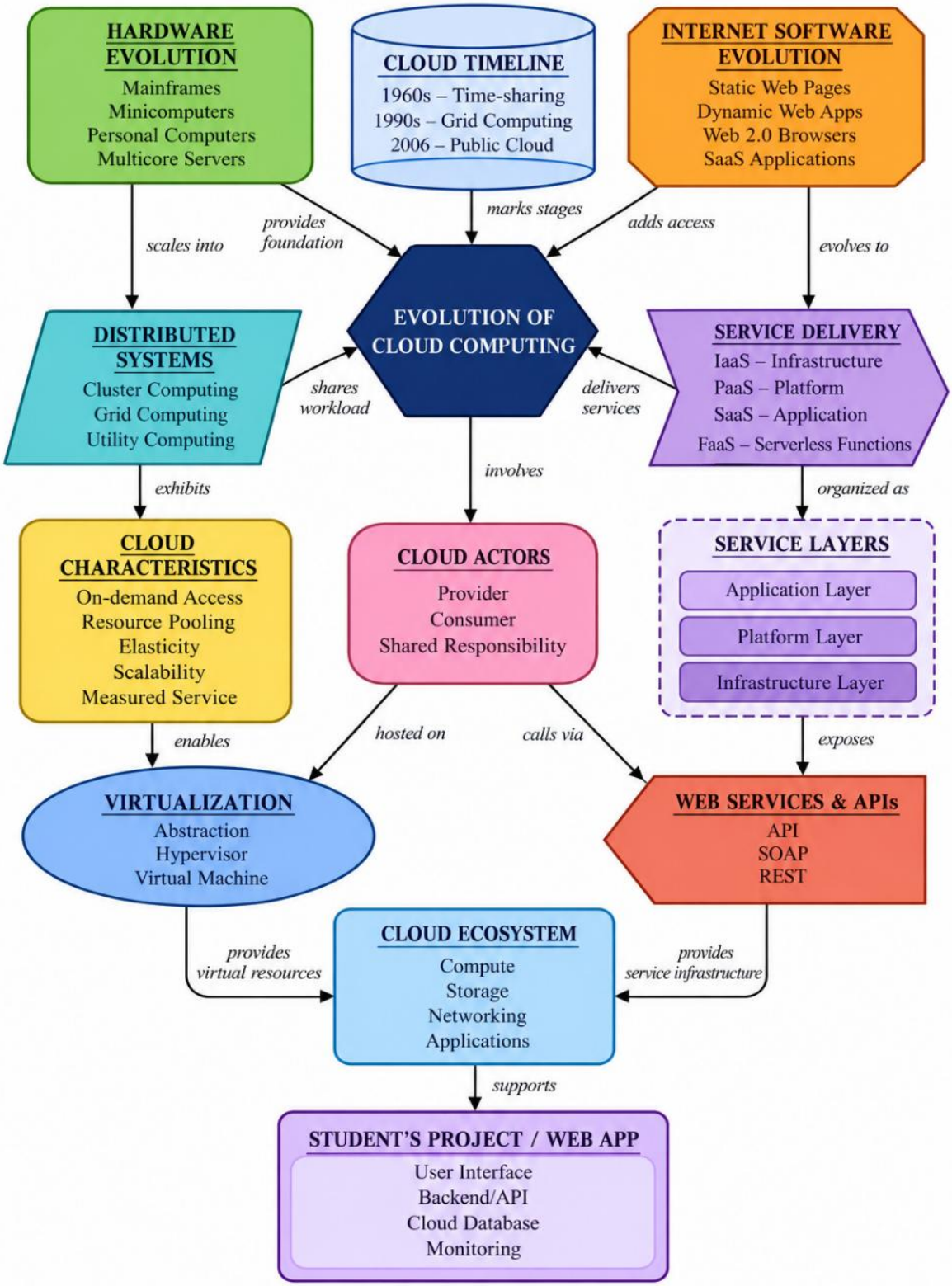
Total Marks:

Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations



1.	Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? The clean visual hierarchy with distinct geometric shapes and directional labeled arrows clearly distinguishes foundational layers from delivery mechanisms, ensuring rapid comprehension of cloud fundamentals.
2.	Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability, elasticity, and resource pooling dynamically match capacity to demand, whereas their absence would cause server crashes during traffic spikes and severe over-provisioning waste during idle periods.
3.	Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? Standardized APIs, REST, and SOAP act as essential communication protocols that enable decoupled system components to seamlessly exchange data and interoperate with minimal latency.
4.	Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? The interconnected relationships between virtualization, web services, and service layers create fault tolerance and optimal load distribution, directly improving system reliability and end-user responsiveness.
5.	Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? Mapping these components highlighted the shift from physical hardware to abstract software-defined ecosystems, and a future redesign could incorporate modern microservices and serverless paradigms for greater granularity.

Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



1.	Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? The concept map clearly connects cloud computing concepts and shows how they support the student project. It simplifies understanding of cloud architecture.
2.	Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability supports growing users, while elasticity adjusts resources based on demand. Without them, performance may decrease and costs may increase.
3.	Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? REST APIs enable communication between frontend and backend. IaaS, PaaS, and SaaS provide infrastructure, development platforms, and ready-to-use applications.
4.	Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? Virtualization, APIs, and cloud services work together to improve resource utilization, communication, reliability, and user experience.
5.	Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? Creating the concept map improved my understanding of cloud architecture. I would improve it by adding security boundaries, automatic scaling, monitoring, and backup mechanisms.

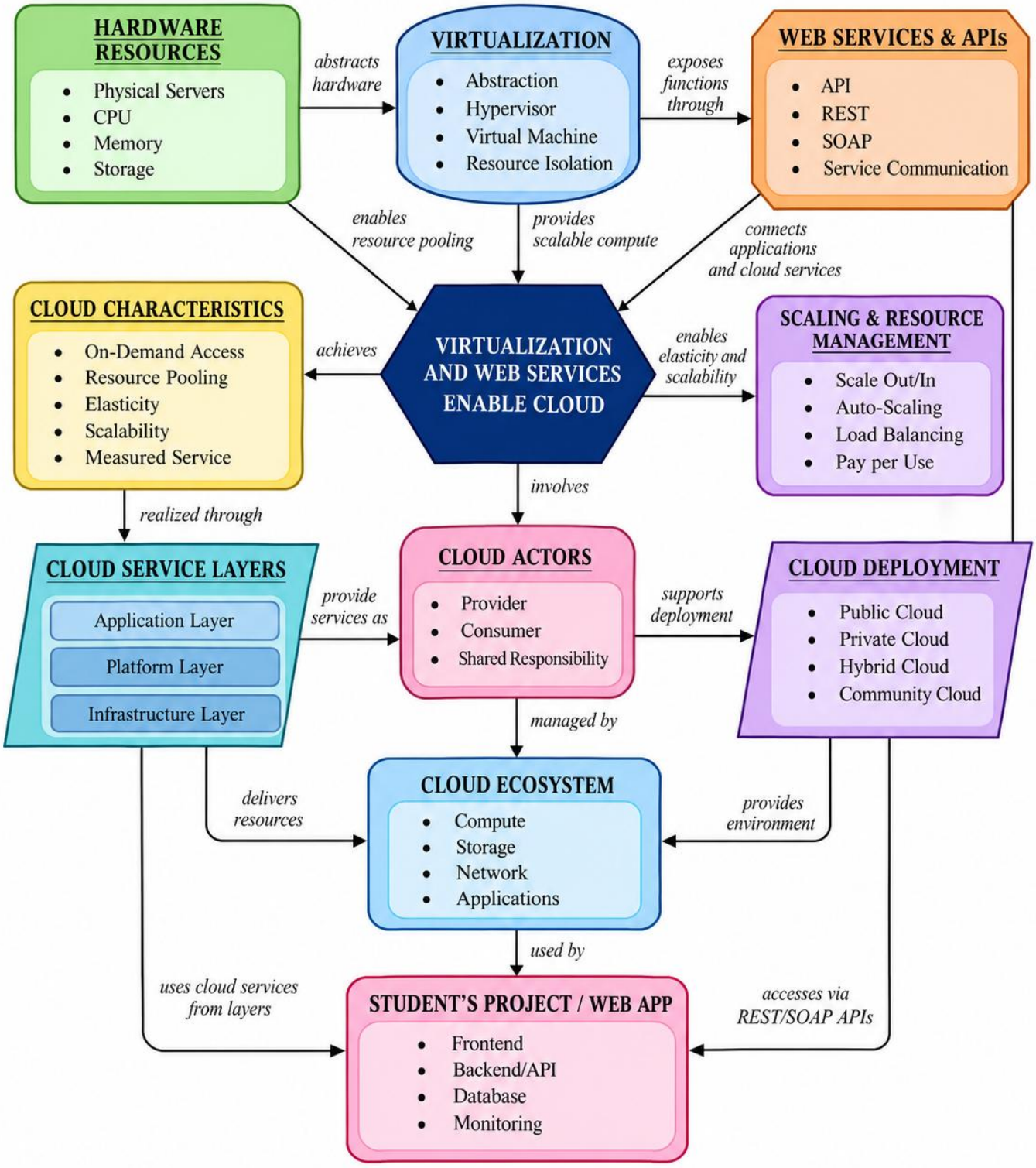
Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
   The concept map clearly connects cloud concepts and service models. Its structured layout helps explain how virtualization, APIs, and cloud actors support cloud applications.

2.	Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability supports increasing users, while elasticity adjusts resources according to demand. Without them, the system may experience slow performance, downtime, and resource wastage.
3.	Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? REST APIs enable efficient communication between applications and cloud services. IaaS provides infrastructure, PaaS supports development, SaaS delivers ready-to-use software, and XaaS offers additional managed services.
4.	Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? Virtualization, APIs, and service models work together to improve resource utilization, communication, and performance. Monitoring and proper resource allocation help improve reliability and user experience.
5.	Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? Creating the concept map improved my understanding of cloud architecture and shared responsibilities. I would improve it by adding security boundaries, automatic scaling, monitoring, and backup mechanisms.

Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



1.	Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? The concept map clearly connects hardware resources, virtualization, APIs, and cloud deployment. These design choices explain how cloud resources support efficient application

	development.
2.	Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? Scalability supports increasing workloads, while elasticity adjusts resources according to demand. Without them, the application may experience slow performance, downtime, and resource wastage.
3.	Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? REST APIs provide lightweight communication, while SOAP supports structured messaging. Virtual machines and managed compute services provide flexible execution environments for cloud applications.
4.	Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? Virtualization enables resource sharing, and APIs connect applications with cloud services. Load balancing, monitoring, and automatic scaling improve performance, availability, and user experience.
5.	Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? Creating the concept map improved my understanding of cloud architecture and shared responsibility. I would improve it by adding detailed security boundaries, API request flows, backup mechanisms, and automatic scaling triggers.